

MIDTERM PROJECT REPORT

Title Page

- **Project Title: Mid-term BUILD AN ADVANCE NETWORK**
- **Course Title: CNIT-134-002**
- **Instructor's Name: Professor Tran**
- **Student's Name: Pao Ze**
- **Date of Submission:**

//This report is submitted in PDF format along with the packet tracer file and a separate configuration file (PDF file), which contains all the commands.

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1. Network Topology

Include the network **topology from Packet Tracer here**, ensuring the following are labeled:

- HQ, Branch Office 1, Branch Office 2
- All routers, switches, servers, wireless devices, and PCs/laptops
- VLANs and IP addressing schemes

2. Network Design and Subnetting Plan

Subnetting Table:

Department	Subnet	Default Gateway	Hosts Supported	First Usable IP	Last Usable IP
IT Department (HQ)	10.0.0.0/25	10.0.0.1	126	10.0.0.1	10.0.0.126
Finance Department (HQ)	10.0.0.192/26	10.0.0.193	62	10.0.0.193	10.0.0.254
HR Department (HQ)	10.0.1.64/27	10.0.1.65	30	10.0.1.65	10.0.1.

Sales Department (BR1)	10.0.1.0/26	10.0.1.1	62	10.0.1.1	10.0.1.62
Support Department (BR1)	10.0.1.96/27	10.0.1.97	30	10.0.1.97	10.0.126
Logistics Department (BR1)	10.0.1.160/27	10.0.1.161	30	10.0.161	10.0.1.190
Marketing Department (BR2)	10.0.0.128/26	10.0.0.129	62	10.0.0.129	10.0.0.190
Research Department (BR2)	10.0.1.192/27	10.0.1.193	30	10.0.1.193	10.0.1.222
Customer Service Department (BR2)	10.0.1.128/27	10.0.1.129	30	10.0.1.129	10.0.158
Servers (HQ)	10.0.1.224/28	10.0.1.225	14	10.0.1.225	10.0.1.238
WLC (HQ)	10.0.1.240/29	10.0.1.241	6	10.0.1.241	10.0.1.246

//All IPs should be displayed in `Courier New` font, using a font size that is one unit smaller.

3. Configuration Steps and Commands

Some notes:

- **IN THIS SECTION, FOCUSING ON EXPLAIN HOW TO DO IT AND WHY YOU DO IT. ADD DEMONSTRATIONS OF THE RESULTS.**
- If there are commands you want to show (recommended), always use `Courier New` font, using a font size that is one unit smaller.
 - For example: `Router(config)#conf t`
- Insert images where appropriate to show the details, **DO NOT ADD TOO MANY.**



Fig. 1. This is a caption of the image

THE FOLLOWING ARE JUST HINTS—BE CREATIVE!

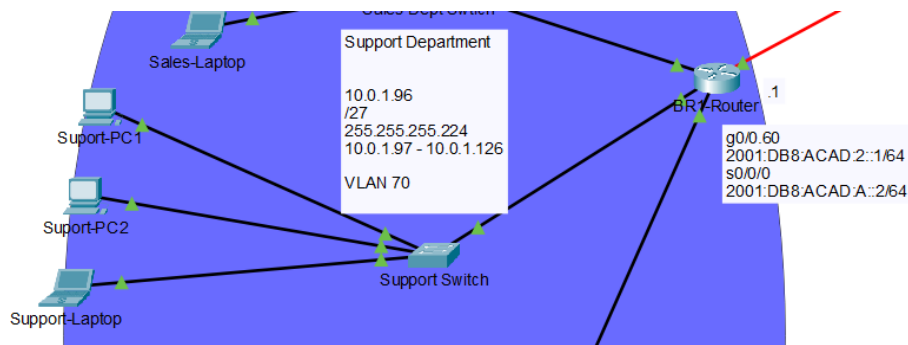
3.1. Basic Network Setup

- Configuration of switches, routers, PCs, and connections.

WHY DO IT: we configure the Switches, routers, and PCs so it can ping other PCs within the same network

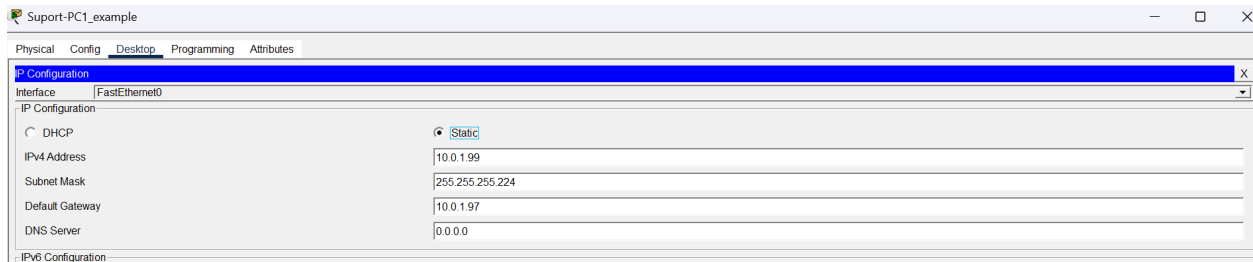
RESULT: You can ping Support-PC1 can ping Support-PC-2

HOW TO DO IT:



- Commands used for device setup (e.g., routing, interfaces, etc.).

Support-PC



Support-Switch

```
Switch>
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 70
Switch(config-vlan)#exit
Switch(config)#int f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 70
Switch(config-if)#int f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 70 ?
<cr>
Switch(config-if)#switchport trunk allowed vlan 70
Switch(config-if)#
```

Router

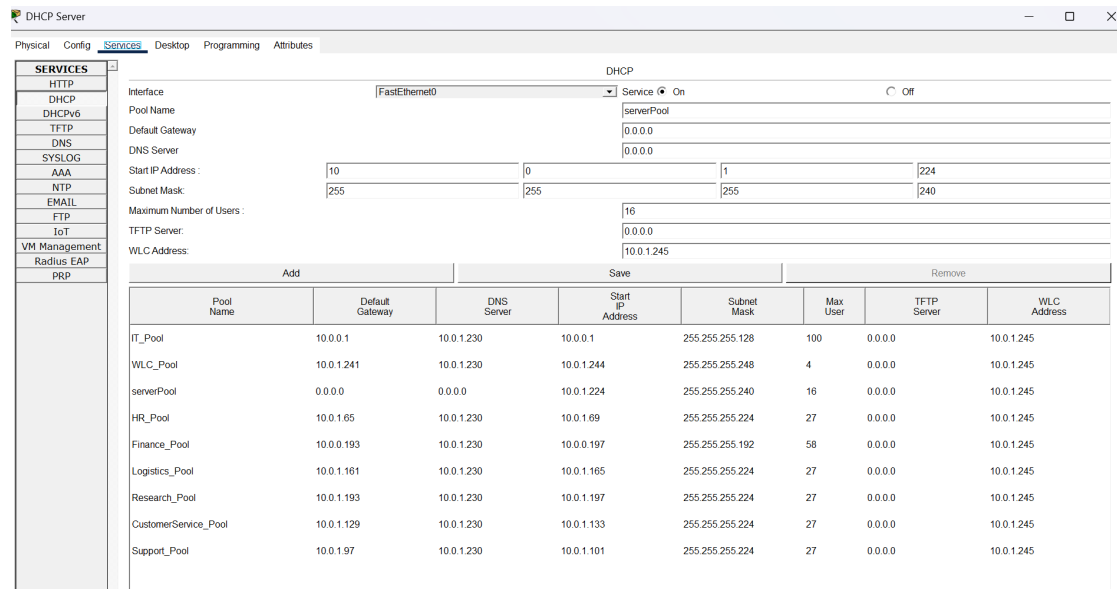
```
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0.70
Router(config-subif)#en
Router(config-subif)#encapsulation do
Router(config-subif)#encapsulation dot1Q 70
Router(config-subif)#ip address 10.0.1.97 255.255.255.224 ?
<cr>
Router(config-subif)#ip address 10.0.1.97 255.255.255.224
Router(config-subif)#int g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.70, changed state to down
```

3.2. DHCP and IPv4/IPv6 Addressing

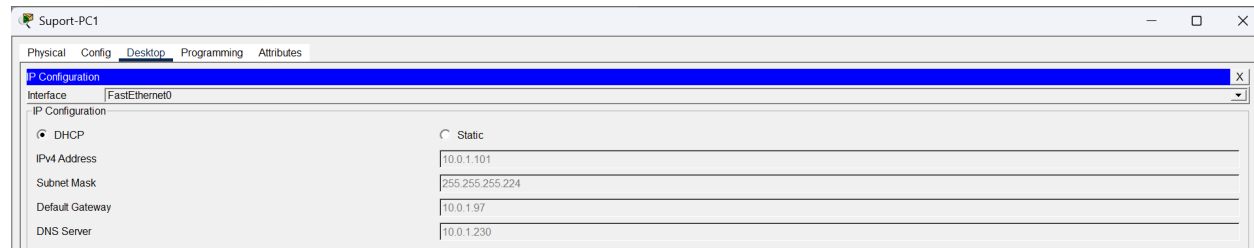
- DHCP pool configurations for each department.



The screenshot shows the DHCP Server configuration window. On the left is a sidebar with a list of services: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. The main area is titled 'DHCP' and shows configuration for the 'FastEthernet0' interface. Below this is a table of DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
IT_Pool	10.0.1	10.0.1.230	10.0.1	255.255.255.128	100	0.0.0.0	10.0.1.245
WLC_Pool	10.0.1.241	10.0.1.230	10.0.1.244	255.255.255.248	4	0.0.0.0	10.0.1.245
serverPool	0.0.0.0	0.0.0.0	10.0.1.224	255.255.255.240	16	0.0.0.0	10.0.1.245
HR_Pool	10.0.1.65	10.0.1.230	10.0.1.69	255.255.255.224	27	0.0.0.0	10.0.1.245
Finance_Pool	10.0.0.193	10.0.1.230	10.0.0.197	255.255.255.192	58	0.0.0.0	10.0.1.245
Logistics_Pool	10.0.1.161	10.0.1.230	10.0.1.165	255.255.255.224	27	0.0.0.0	10.0.1.245
Research_Pool	10.0.1.193	10.0.1.230	10.0.1.197	255.255.255.224	27	0.0.0.0	10.0.1.245
CustomerService_Pool	10.0.1.129	10.0.1.230	10.0.1.133	255.255.255.224	27	0.0.0.0	10.0.1.245
Support_Pool	10.0.1.97	10.0.1.230	10.0.1.101	255.255.255.224	27	0.0.0.0	10.0.1.245

- Some examples of IP configurations on Routers, PCs, servers.



The screenshot shows the Support-PC1 configuration window. The 'IP Configuration' tab is selected. It shows the configuration for the 'FastEthernet0' interface. The 'DHCP' radio button is selected, and the 'Static' radio button is unselected. Below this, the IP configuration details are shown.

Interface	IP Address	Subnet Mask	Default Gateway	DNS Server
FastEthernet0	10.0.1.101	255.255.255.224	10.0.1.97	10.0.1.230



- VLAN creation and port assignment for each department.



- Configuration of trunk ports and inter-VLAN routing using subinterfaces.

```
Support_SW#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50,60,70,80,90,100,110

Port      Vlans allowed and active in management domain
Fa0/1     70

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     70

Support_SW#|
```

3.4. FHRP Configuration

- HSRP virtual IPs and priorities are correctly set up for each VLAN.

HQ-Router 1	HQ-Router 2
<pre>Physical Config CLI Attributes ! interface GigabitEthernet0/0 no ip address duplex auto speed auto ! interface GigabitEthernet0/0.10 encapsulation dot1Q 10 ip address 10.0.0.2 255.255.255.128 ip helper-address 10.0.1.228 standby 10 ip 10.0.0.1 standby 10 priority 110 standby 10 preempt ! interface GigabitEthernet0/0.20 encapsulation dot1Q 20 ip address 10.0.0.194 255.255.255.192 ip helper-address 10.0.1.228 standby 20 ip 10.0.0.193 standby 20 priority 110 standby 20 preempt ! interface GigabitEthernet0/0.30 encapsulation dot1Q 30 ip address 10.0.1.66 255.255.255.224 ip helper-address 10.0.1.228 ip access-group BOBBY in ipv6 address 2001:DB8:ACAD:1::2/64 standby 30 ip 10.0.1.65 standby 30 priority 110 standby 30 preempt ! interface GigabitEthernet0/0.40 encapsulation dot1Q 40 ip address 10.0.1.226 255.255.255.240 ipv6 address 2001:DB8:ACAD:4::2/64 standby 40 ip 10.0.1.225 standby 40 priority 110 standby 40 preempt ! interface GigabitEthernet0/0.50 encapsulation dot1Q 50 ip address 10.0.1.242 255.255.255.248 ip helper-address 10.0.1.228 standby 50 ip 10.0.1.241 standby 50 priority 110 standby 50 preempt !</pre>	<pre>Physical Config CLI Attributes ! interface GigabitEthernet0/0 no ip address duplex auto speed auto ! interface GigabitEthernet0/0.10 encapsulation dot1Q 10 ip address 10.0.0.3 255.255.255.128 ip helper-address 10.0.1.228 standby 10 ip 10.0.0.1 standby 10 priority 120 standby 10 preempt ! interface GigabitEthernet0/0.20 encapsulation dot1Q 20 ip address 10.0.0.195 255.255.255.192 ip helper-address 10.0.1.228 standby 20 ip 10.0.0.193 standby 20 priority 120 standby 20 preempt ! interface GigabitEthernet0/0.30 encapsulation dot1Q 30 ip address 10.0.1.67 255.255.255.224 ip helper-address 10.0.1.228 ip access-group COLT in ipv6 address 2001:DB8:ACAD:1::3/64 standby 30 ip 10.0.1.65 standby 30 priority 120 standby 30 preempt ! interface GigabitEthernet0/0.40 encapsulation dot1Q 40 ip address 10.0.1.227 255.255.255.240 ipv6 address 2001:DB8:ACAD:4::3/64 standby 40 ip 10.0.1.225 standby 40 priority 120 standby 40 preempt ! interface GigabitEthernet0/0.50 encapsulation dot1Q 50 ip address 10.0.1.243 255.255.255.248 ip helper-address 10.0.1.228 standby 50 ip 10.0.1.241 standby 50 priority 120 standby 50 preempt !</pre>

- Active and standby roles are clearly identified.

3.5. Routing Configuration

- OSPF dynamic routing setup in Area 0.
- Static routes for IPv6.

BR1-Router	BR2-Router
Physical Config <u>CLI</u> Attributes <pre> interface Serial0/0/1 no ip address clock rate 2000000 shutdown ! interface Vlan1 no ip address shutdown ! router ospf 1 router-id 3.3.3.3 log-adjacency-changes passive-interface GigabitEthernet0/0.60 passive-interface GigabitEthernet0/1.70 passive-interface GigabitEthernet0/2.80 network 10.0.1.0 0.0.0.63 area 0 network 10.0.1.96 0.0.0.31 area 0 network 10.0.1.160 0.0.0.31 area 0 network 198.51.100.0 0.0.0.3 area 0 ! ip classless ! ip flow-export version 9 ! ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:A::1 ipv6 route 2001:DB8:ACAD:4::/64 2001:DB8:ACAD:A::1 ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:A::1 ! ip bgp-as-list extended BGP AS </pre>	Physical Config <u>CLI</u> Attributes <pre> clock rate 2000000 shutdown ! interface Vlan1 no ip address shutdown ! router ospf 1 router-id 4.4.4.4 log-adjacency-changes passive-interface GigabitEthernet0/0.90 passive-interface GigabitEthernet0/1.100 passive-interface GigabitEthernet0/2.110 network 10.0.0.128 0.0.0.63 area 0 network 10.0.1.128 0.0.0.31 area 0 network 10.0.1.192 0.0.0.31 area 0 network 198.51.100.4 0.0.0.3 area 0 ! ip classless ! ip flow-export version 9 ! ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:B::1 ipv6 route 2001:DB8:ACAD:4::/64 2001:DB8:ACAD:B::1 ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:B::1 ! </pre>
HQ-Router 1 Physical Config <u>CLI</u> Attributes <pre> no ip address clock rate 2000000 shutdown ! interface Vlan1 no ip address shutdown ! router ospf 1 router-id 1.1.1.1 log-adjacency-changes network 10.0.0.0 0.0.0.127 area 0 network 10.0.0.192 0.0.0.63 area 0 network 10.0.1.64 0.0.0.31 area 0 network 10.0.1.224 0.0.0.15 area 0 network 10.0.1.240 0.0.0.7 area 0 network 198.51.100.8 0.0.0.3 area 0 ! ip classless ! ip flow-export version 9 ! ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:C::1 ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:C::1 ! </pre>	HQ-Router 2 Physical Config <u>CLI</u> Attributes <pre> no ip address clock rate 2000000 shutdown ! interface Vlan1 no ip address shutdown ! router ospf 1 router-id 2.2.2.2 log-adjacency-changes network 10.0.0.0 0.0.0.127 area 0 network 10.0.0.192 0.0.0.63 area 0 network 10.0.1.64 0.0.0.31 area 0 network 10.0.1.224 0.0.0.15 area 0 network 10.0.1.240 0.0.0.7 area 0 network 198.51.100.12 0.0.0.3 area 0 ! ip classless ! ip flow-export version 9 ! ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:D::1 ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:D::1 ! </pre>

Central Router

Physical Config **CLI** Attributes

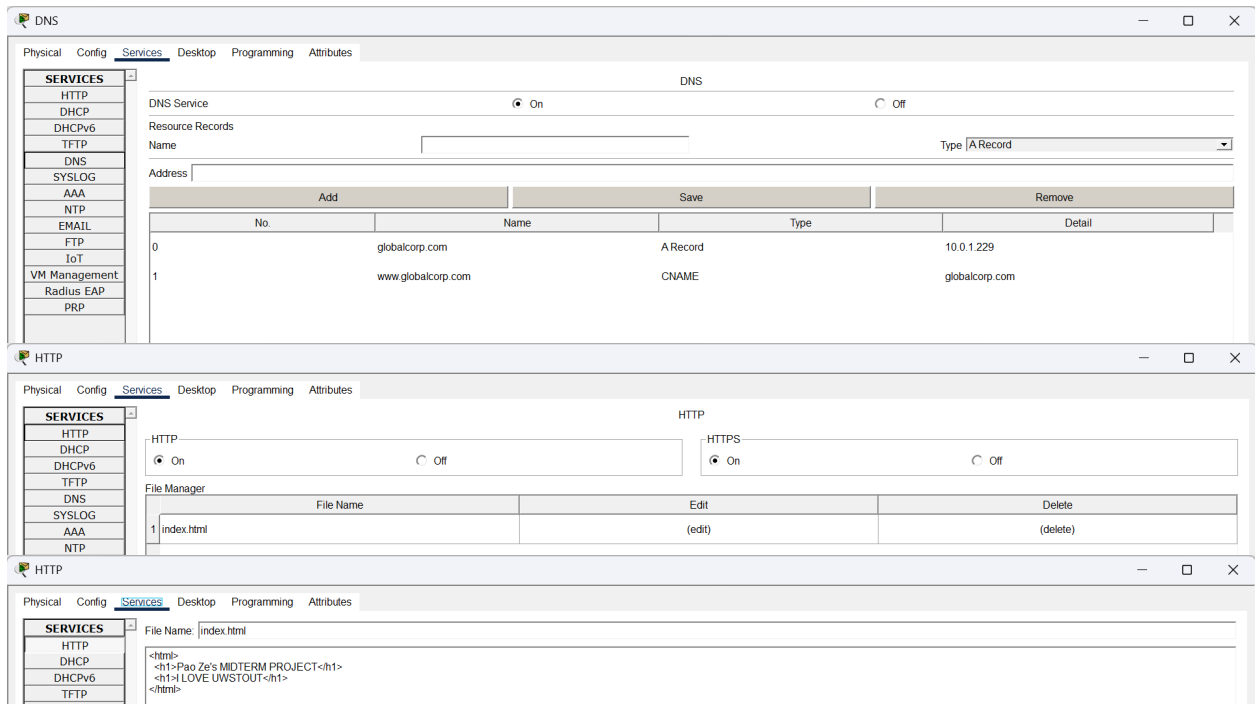
```

ipv6 address 2001:DB8:ACAD:C::1/64
clock rate 2000000
!
interface Serial0/1/1
ip address 198.51.100.14 255.255.255.252
ipv6 address 2001:DB8:ACAD:D::1/64
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 5.5.5.5
log-adjacency-changes
network 198.51.100.0 0.0.0.3 area 0
network 198.51.100.4 0.0.0.3 area 0
network 198.51.100.8 0.0.0.3 area 0
network 198.51.100.12 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
ipv6 route 2001:DB8:ACAD:1::/64 2001:DB8:ACAD:C::2
ipv6 route 2001:DB8:ACAD:4::/64 2001:DB8:ACAD:C::2
ipv6 route 2001:DB8:ACAD:2::/64 2001:DB8:ACAD:A::2
ipv6 route 2001:DB8:ACAD:3::/64 2001:DB8:ACAD:B::2
!
!

```

3.6. Other Server Configuration

- DNS, HTTP/HTTPS, and FTP servers are configured with correct addressing.
- Services (e.g., file permissions, domain records) are set as required.
- All services are accessible to authorized devices and users.



The image shows three screenshots of a network management interface for configuring services.

DNS Configuration: The 'DNS' window shows the 'SERVICES' tab with 'DNS' selected. The 'DNS Service' is set to 'On'. Below, the 'Resource Records' table is visible:

No.	Name	Type	Detail
0	globalcorp.com	A Record	10.0.1.229
1	www.globalcorp.com	CNAME	globalcorp.com

HTTP Configuration: The 'HTTP' window shows the 'SERVICES' tab with 'HTTP' selected. The 'HTTP' service is set to 'On'. Below, the 'File Manager' table is visible:

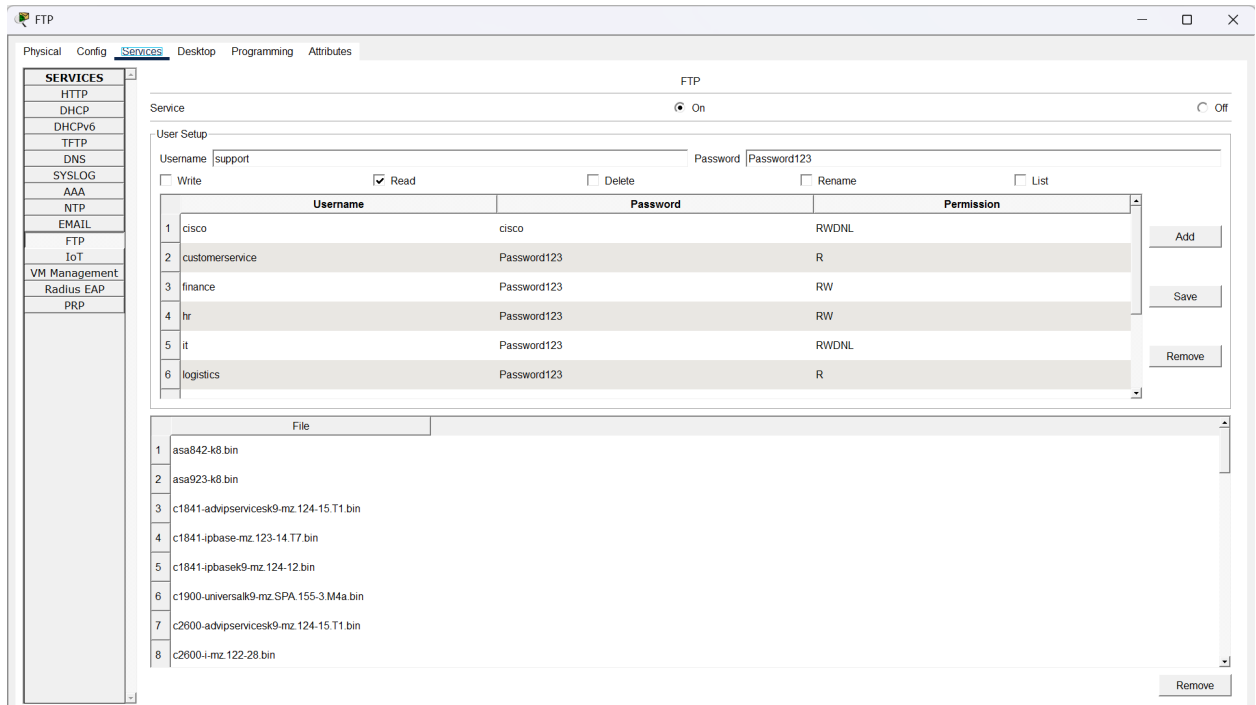
File Name	Edit	Delete
1 index.html	(edit)	(delete)

HTTP Content: A third window shows the content of the 'index.html' file:

```

<html>
<h1>Pao Ze's MIDTERM PROJECT</h1>
<h1>I LOVE UWSTOUT</h1>
</html>

```



FTP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

Service: **FTP** On

User Setup

Username: support Password: Password123

☐ Write ☒ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	customerservice	Password123	R
3	finance	Password123	RW
4	hr	Password123	RW
5	it	Password123	RWDNL
6	logistics	Password123	R

Buttons: Add, Save, Remove

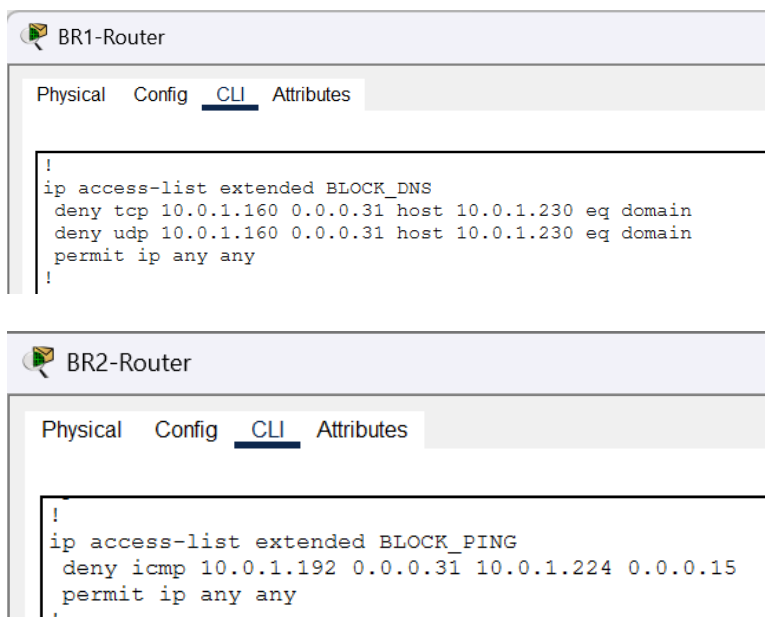
File

- asa842-k8.bin
- asa923-k8.bin
- c1841-adwpservicesk9-mz.124-15.T1.bin
- c1841-ipbase-mz.123-14.T7.bin
- c1841-ipbasek9-mz.124-12.bin
- c1900-universalk9-mz.SPA.155-3.M4a.bin
- c2600-adwpservicesk9-mz.124-15.T1.bin
- c2600-i-mz.122-28.bin

Buttons: Remove

3.7. Access Control Lists

- Configuration and testing ACLs.



BR1-Router

Physical Config **CLI** Attributes

```

!
ip access-list extended BLOCK_DNS
deny tcp 10.0.1.160 0.0.0.31 host 10.0.1.230 eq domain
deny udp 10.0.1.160 0.0.0.31 host 10.0.1.230 eq domain
permit ip any any
!

```

BR2-Router

Physical Config **CLI** Attributes

```

!
ip access-list extended BLOCK_PING
deny icmp 10.0.1.192 0.0.0.31 10.0.1.224 0.0.0.15
permit ip any any
!

```

HQ-Router 1

Physical Config CLI Attributes

```
!  
ip access-list extended BOBBY  
deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq www  
deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq 443  
permit ip any any  
!
```

HQ-Router 2

Physical Config CLI Attributes

```
!  
ip access-list extended COLT  
deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq www  
deny tcp 10.0.1.64 0.0.0.31 host 10.0.1.229 eq 443  
permit ip any any  
!
```

3.8. Wireless Network and RADIUS Setup

- WLC setup for centralized management.
- Configuration of SSIDs.
- Configuration of Lightweight APs.
- **RADIUS server setup for authentication.[NOT NEEDED]**

WLC

Physical Config Attributes

GLOBAL Settings

INTERFACE

GigabitEthernet1

GigabitEthernet2

GigabitEthernet3

GigabitEthernet4

GigabitEthernet5

Management

Management

IP Configuration

IPv4 Address 10.0.1.245

Subnet Mask 255.255.255.248

Default Gateway 10.0.1.241

DNS Server 10.0.1.230

IT Access Point

Physical Config Attributes

GLOBAL Settings

WLC

INTERFACE

Primary Controller 10.0.1.245

IT Access Point

Physical Config Attributes

GLOBAL Settings

Display Name IT Access Point

Gateway/DNS IPv4

☒ DHCP

☐ Static

Default Gateway 10.0.0.1

DNS Server 10.0.1.230

Device Clock 04:16:12 Mon Mar 1 1993 UTC

IT-PC1

Physical Config Desktop Programming Attributes

Web Browser

URL: https://10.0.1.245/frameMonitor.html

Go Stop

Sage Configuration Ping Logout Refresh Home

MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

Monitor

Summary

150 Access Points Supported



Controller Summary

Management IP Address	192.168.1.1, 11/128
Software Version	8.3.111.0
Field Recovery Image Version	7.6.101.1
System Name	WLC
Up Time	17 minutes, 54 seconds
System Time	Wed Apr 1 21:41:59 2026
Redundancy Mode	N/A
Internal Temperature	+31 C
802.11s Network State	Enabled
802.11b/g Network State	Enabled
Local Mobility Group	
CPU(s) Usage	0%
Individual CPU Usage	0%/1%, 0%/0%
Memory Usage	46%
Fan Status	3800 rpm

Access Point Summary

	Total	Up	Down	
802.11a/n/ac Radios	3	3	0	Detail
802.11b/g/n Radios	3	3	0	Detail
Dual-Band Radios	0	0	0	Detail
All APs	3	3	0	Detail

Client Summary

Current Clients	3	Detail
Excluded Clients	0	Detail
Disabled Clients	0	Detail

Top

IT-PC1

Physical Config Desktop Programming Attributes

Web Browser

URL: https://10.0.1.245/frameWlan.html

Go Stop

Sage Configuration Ping Logout Refresh Home

MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

WLANs

WLANs

Advanced

AP Groups

Current Filter: [Change Filter] [Clear Filter]

Create New Go

Entries 1 - 4 of 4

WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies	
1	WLAN	test@123	test@123	Enabled	[WPA2][Auth(PSK)]	Remove
2	WLAN	WIFI_FT	WIFI_FT	Enabled	[WPA2][Auth(PSK)]	Remove
3	WLAN	WIFI_Finance	WIFI_Finance	Enabled	[WPA2][Auth(PSK)]	Remove
4	WLAN	WIFI_HR	WIFI_HR	Enabled	[WPA2][Auth(PSK)]	Remove

IT-PC1

Physical Config Desktop Programming Attributes

Web Browser

URL: https://10.0.1.245/frameWireless.html

Go Stop

Sage Configuration Ping Logout Refresh Home

MONITOR WLANs CONTROLLER WIRELESS SECURITY MANAGEMENT COMMANDS HELP FEEDBACK

Wireless

All APs

Current Filter: [Change Filter] [Clear Filter]

Number of APs: 3

AP Name	IP Address (IPv4/IPv6)	AP Model	AP MAC	AP Up Time	Admin Status	Operational Status	PoE Status
Finance Access Point	10.0.0.201	AIR-CT5502-K9	00:0A:F3:19:53:01	0 d, 0 h 18 m 59 s	Enabled	REG	-
HR Access Point	10.0.1.71	AIR-CT5502-K9	00:90:0C:AE:4B:01	0 d, 0 h 18 m 59 s	Enabled	REG	-
IT Access Point	10.0.0.5	AIR-CT5502-K9	00:0B:BE:91:B5:01	0 d, 0 h 18 m 59 s	Enabled	REG	-

4. Testing and Verification

- Ping Tests:** Document results of pings within and between branches. Test with both IPv4 and IPv6

```
Sale-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 29ms, Average = 12ms
C:\>ping 198.51.100.5
Pinging 198.51.100.5 with 32 bytes of data:
Reply from 198.51.100.5: bytes=32 time=33ms TTL=253
Reply from 198.51.100.5: bytes=32 time=2ms TTL=253
Reply from 198.51.100.5: bytes=32 time=3ms TTL=253
Reply from 198.51.100.5: bytes=32 time=2ms TTL=253
Ping statistics for 198.51.100.5:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 33ms, Average = 10ms
C:\>ping 10.0.0.129
Pinging 10.0.0.129 with 32 bytes of data:
Reply from 10.0.0.129: bytes=32 time=33ms TTL=253
Reply from 10.0.0.129: bytes=32 time=17ms TTL=253
Reply from 10.0.0.129: bytes=32 time=4ms TTL=253
Reply from 10.0.0.129: bytes=32 time=4ms TTL=253
Ping statistics for 10.0.0.129:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 4ms, Maximum = 33ms, Average = 14ms
C:\>ping 10.0.0.130
Pinging 10.0.0.130 with 32 bytes of data:
Request timed out.
Reply from 10.0.0.130: bytes=32 time=17ms TTL=125
Reply from 10.0.0.130: bytes=32 time=2ms TTL=125
Reply from 10.0.0.130: bytes=32 time=22ms TTL=125
Ping statistics for 10.0.0.130:
  Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 22ms, Average = 13ms
C:\>clear
Invalid Command.
C:\>
```

- **Traceroute Analysis:** Include screenshots or `tracert` results.

```
C:\>tracert 10.0.0.130

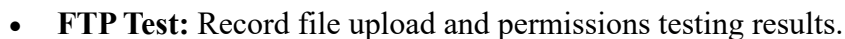
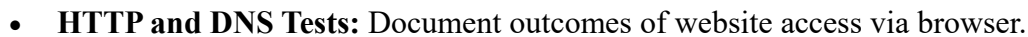
Tracing route to 10.0.0.130 over a maximum of 30 hops:

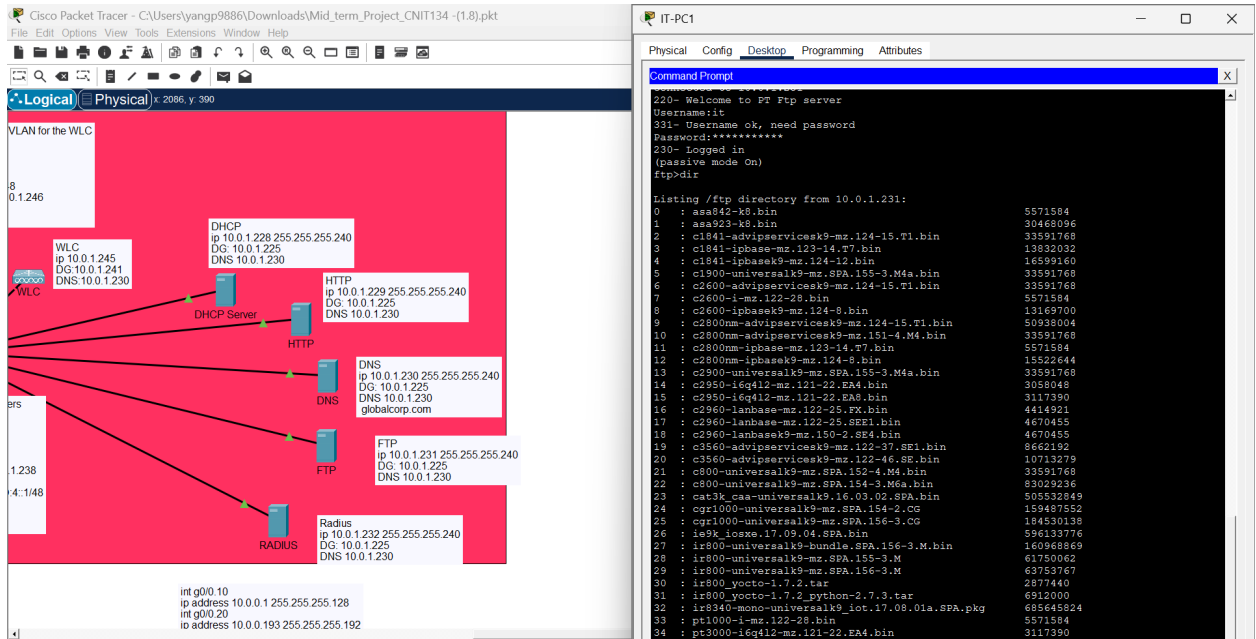
  0  0 ms    0 ms    0 ms    10.0.1.1
  1  1 ms    1 ms    1 ms    198.51.100.2
  2  1 ms    1 ms    2 ms    198.51.100.5
  3  0 ms    1 ms    1 ms    10.0.0.130

Trace complete.

C:\>
```

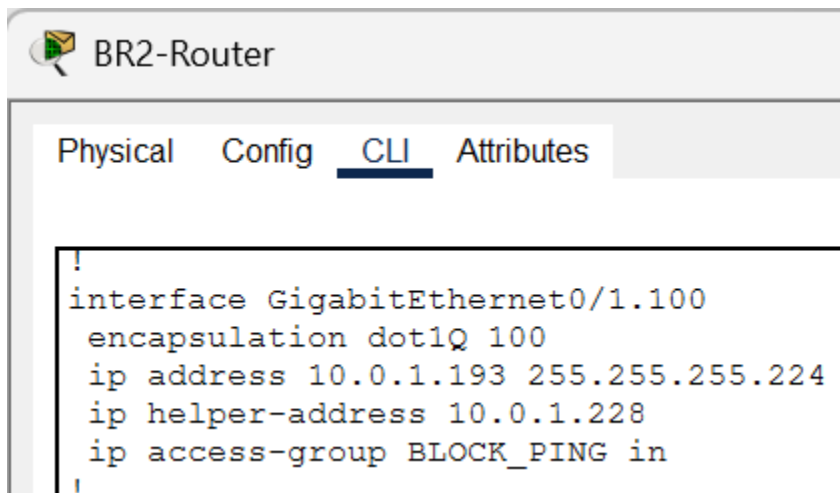
- **HSRP Test:** Connectivity remains uninterrupted during failover.





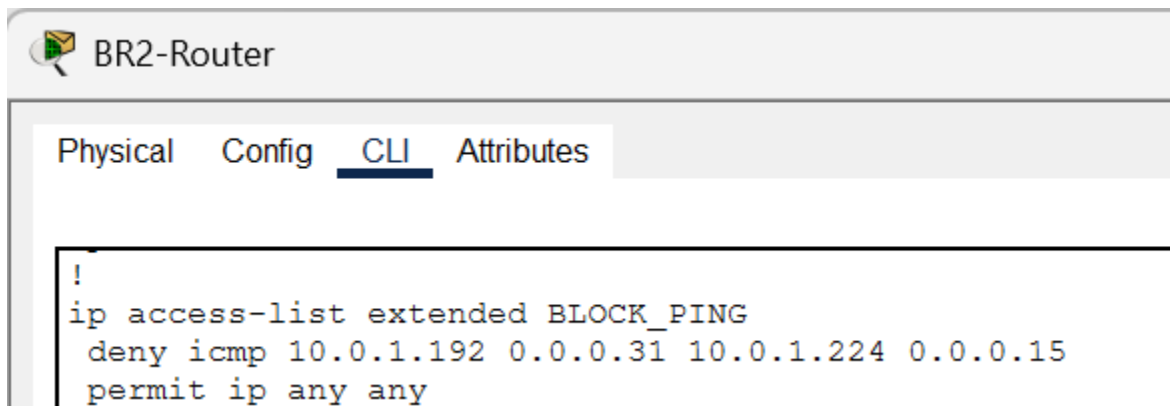
The image shows a Cisco Packet Tracer network diagram and a command prompt window. The network diagram features a central router (R1) connected to a switch (S1) and a server (S2). The switch is configured with a VLAN for the WLC (Wireless LAN Controller) and is connected to a DHCP Server, HTTP, DNS, FTP, and Radius. The server is connected to the switch and has a configuration for the WLC. The command prompt window shows the output of the 'show ip dir' command, listing various files and their sizes.

- **ACL Test:** Show the effects of ACL lists



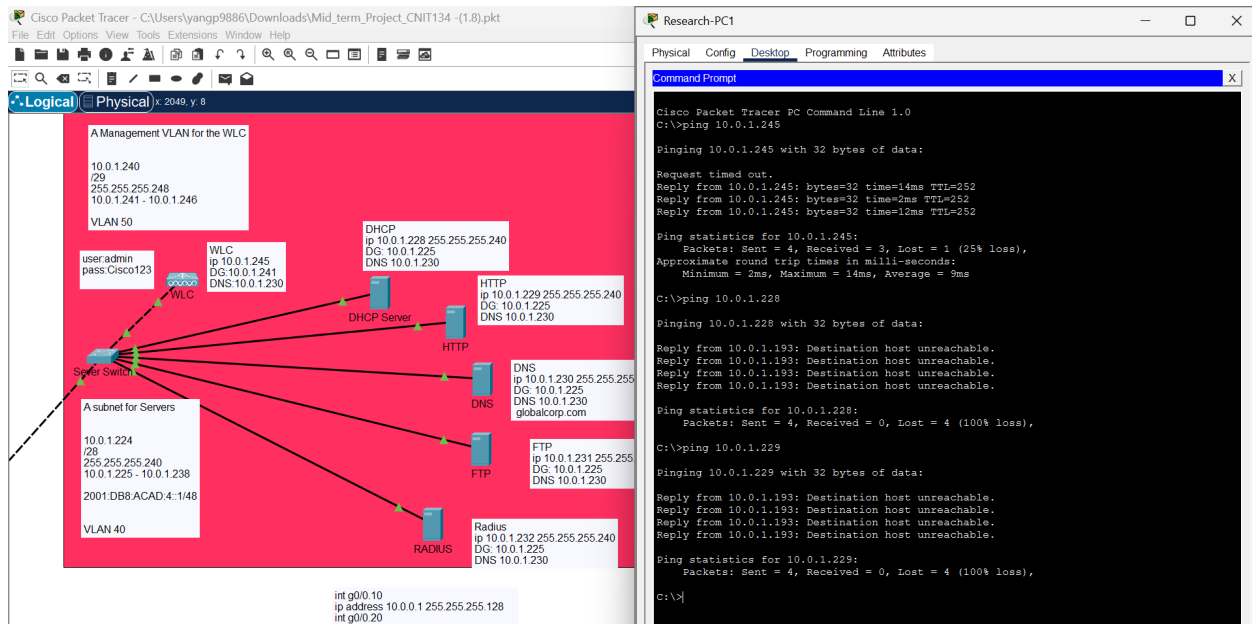
The image shows the CLI configuration for the BR2-Router. The configuration includes the following commands:

```
interface GigabitEthernet0/1.100
 encapsulation dot1Q 100
 ip address 10.0.1.193 255.255.255.224
 ip helper-address 10.0.1.228
 ip access-group BLOCK_PING in
```



The image shows the CLI configuration for the BR2-Router. The configuration includes the following commands:

```
ip access-list extended BLOCK_PING
 deny icmp 10.0.1.192 0.0.0.31 10.0.1.224 0.0.0.15
 permit ip any any
```



- **Wireless Network Tests:** Connectivity tests and authentication tests.

5. Challenges and Solutions

- Outline issues encountered during the project.

My most issues when it comes to this project was the WLC, because we only did only one assignment in class. My experience around doing it was very small.

- Provide solutions implemented to resolve them.

My solution was to ask for help, as if I couldn't do it I needed someone more informed explain how I had to fix it. Otherwise I had the WLC and Access Points exactly how they needed to be

6. Conclusion

- **Summary of the project's success.**

This midterm project successfully demonstrated the design and implementation of an advanced enterprise network spanning three sites: a headquarters (HQ) and two branch offices (BR1 and BR2). The network was built to support nine departments across 11 subnets, with each subnet carefully sized using variable-length subnet masking (VLSM) to efficiently allocate IP address space. All major enterprise networking technologies were configured and verified, including VLANs, inter-VLAN routing via subinterfaces, DHCP, OSPF dynamic routing, HSRP for gateway redundancy, ACLs for traffic control, DNS and HTTP/FTP servers, and a centralized wireless infrastructure managed through a Wireless LAN Controller (WLC). The network passed connectivity testing through ping and

traceroute verification, HSRP failover testing, and wireless authentication tests. Overall, the project met its objectives and resulted in a functional, scalable network that reflects a real-world enterprise environment.

- **Key learnings from the project.**

- Working through this project provided valuable hands-on experience with technologies that are essential in modern network administration. Subnetting and IP planning required careful thought to ensure that each department received the right number of usable addresses without wasting space. Configuring VLANs and trunk ports reinforced how traffic segmentation improves both security and performance in a multi-department environment. Setting up OSPF across multiple sites highlighted how dynamic routing protocols simplify route management compared to manually maintaining static routes. HSRP configuration showed how redundancy can be built into a network so that a single router failure does not interrupt service for end users. The ACL configuration section deepened an understanding of how to control traffic flow at the network boundary and between segments. Perhaps the most challenging and instructive part of this project was configuring the Wireless LAN Controller and lightweight access points. Since classroom exposure to WLC was limited, completing this section required seeking guidance and working through problems independently, which ultimately built confidence in troubleshooting unfamiliar technology. Overall, this project brought together nearly every topic covered in CNIT 134 into a single cohesive network, reinforcing how these technologies work together in a production-like environment.